AI for Economic Research: Dynamic Models, Language, and Agents
Lecture 7.1: Text as Economic Measurement

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Lecture 7 in One Map

you are here ▼

7.1 FRAME
text as measurement
what generates this text?

7.2 REPRESENT
tokens $\longrightarrow$ vectors
how do we turn it into numbers?

7.3 ATTEND
attention & Transformer
how does context get encoded?

7.4 PRETRAIN
LMs & alignment
where does a pretrained model come from?

7.5 MEASURE
limits, apps, validation
is my measure valid — and defensible?

WHY TEXT = MEASUREMENT | **HOW THE MACHINE READS TEXT** | **MEASURE, VALIDATE, DISCLOSE**

the spine: $x_i = g(s_i, a_i, c_i) + \varepsilon_i \longrightarrow m(x_i) \longrightarrow y_i = \beta\, m(x_i) + \gamma z_i + u_i$

before 7.1: Lec03–04 gave you neural nets, back-prop & PyTorch — Lec07 teaches them to read

after 7.5: Lec08 RAG (grounded memory) $\rightarrow$ Lec09 agents (grounded action) $\rightarrow$ Lec10 cases

Two case threads run through all five acts — **Thread A: the EPU index** *(Baker–Bloom–Davis 2016), the transparent dictionary baseline every method must beat;* **Thread B: FOMC hawk–dove** *(central-bank communication), the running example for embeddings, attention, and validation. Watch for them.*

Text as Data

Why language became an economic measurement problem

The Opening Hook: Reading Uncertainty out of Newspapers

In 2016, three economists asked: can you read policy uncertainty out of the newspaper?

- Baker, Bloom & Davis (2016, *QJE*): scan 10 major US newspapers, month by month, and count articles containing **all three** of {`economic`/`economy`}, **and** {`uncertain`/`uncertainty`}, **and** one of {`Congress`, `deficit`, `Federal Reserve`, `legislation`, `regulation`, `White House`}.
- Scale by each paper's total output, standardize, average across papers, index to 100:

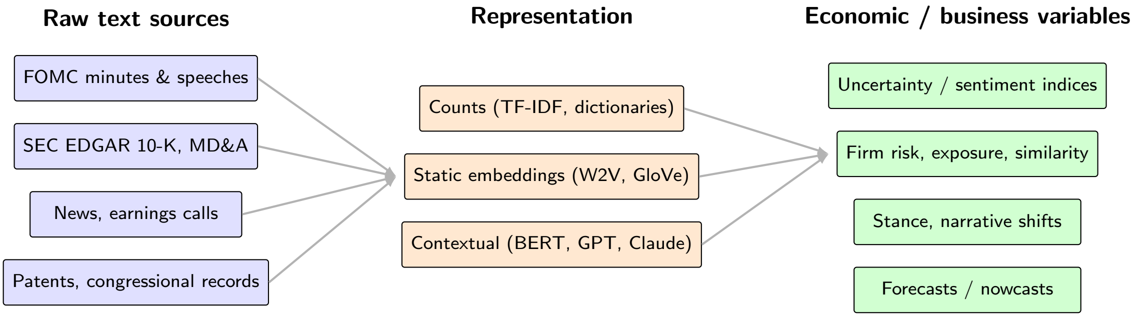
$$\mathrm{EPU}_\tau \;=\; 100 \times \frac{u_\tau}{\bar{u}}, \qquad u_\tau \;=\; \frac{1}{10}\sum_{p=1}^{10} \frac{c_{p\tau}/N_{p\tau}}{\sigma_p}.$$

  where $p = 1, \ldots, 10$ indexes newspapers, τ the month; $c_{p\tau}$ = qualifying articles in paper p; $N_{p\tau}$ = paper p's total articles (dividing removes size differences); σ_p = std. dev. of paper p's scaled series (so no one paper dominates); u_τ = 10-paper average; $\bar{u}$ = its sample mean, so $\times 100$ sets the average to 100.

- No statistical agency publishes "policy uncertainty" — it is not directly observed anywhere. Yet in a VAR, EPU spikes precede investment and employment declines.

The move that matters: unstructured language became a *regression-ready variable* — this act asks what that takes, and where it goes wrong. *(Thread A begins.)*

Text as Economic Data — The Big Picture



The lecture builds the middle column rung by rung: Act 2 counts → static embeddings, Act 3 contextual machinery, Act 4 how those models are trained — and Act 5 turns the machinery into the right-hand column, with validation.

Why Economists Care About LLMs

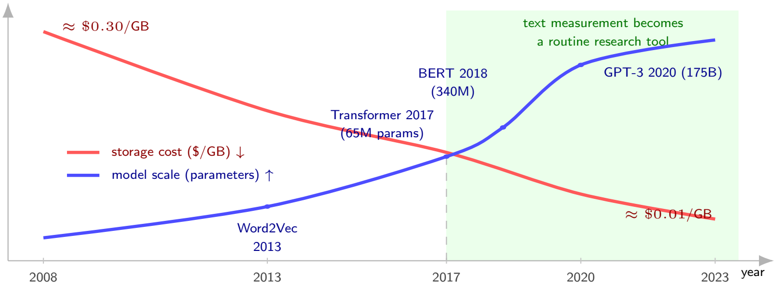
- **Text as data —** modern economic research increasingly leverages textual sources:
  - Corporate filings (10-K, 10-Q, MD&A)
  - Central bank speeches, FOMC minutes, ECB statements
  - Earnings calls, analyst reports, regulatory filings
  - News, social media, congressional records
- **Applications:**
  - Sentiment analysis for financial forecasting
  - Policy stance detection and uncertainty indices
  - New text-derived macroeconomic indicators
  - Summarization of long, technical documents
- **Why LLMs in particular:**
  - Capture nuanced language and domain-specific jargon
  - Handle long-range context (sentences, paragraphs, whole documents)
  - Provide a unified architecture for many tasks

Three Roles of Text in Economic Research

Role	What text is	Example
Measurement tool	A way to construct previously unobserved variables	Baker–Bloom–Davis (2016) Economic Policy Uncertainty index from newspapers
Behavioral outcome	A response variable revealing decisions or institutional change	Hansen, McMahon & Prat (2018): how FOMC transparency reform changed committee speech
Treatment variable	A quasi-experimental input affecting behavior	Central bank communication shocks and inflation expectations

Why Now? The Feasibility Window

Two curves crossed around 2017: the cost of handling text collapsed while model capability exploded.



Both quantities on log scales; curves are stylized interpolations through the marked real data points. Storage costs after McCallum's disk-price series; model sizes from Vaswani et al. (2017), Devlin et al. (2018), Brown et al. (2020).

The Spine

A data-generating process for text — and the two hazards it creates

An Identification-First View of Text

- **Engineering perspective first:** to a machine, a document is just a character string. But every string was *produced* — by a person, in a situation, for an audience. Before any econometrics, ask: what generated this string?

- **The data-generating process (DGP):**

$$x_i \;=\; g\big(s_i,\; a_i,\; c_i\big) \;+\; \varepsilon_i$$

  - s_i: the latent **economic state** we want to measure (policy stance, sentiment, firm risk)
  - a_i: the speaker's **strategic framing** — audience management, spin, emphasis, tone chosen for effect
  - c_i: **institutional context** — venue, legal disclosure requirements, format conventions

- **Why this matters for a downstream regression** $y_i = \beta\, m(x_i) + \gamma z_i + u_i$: if the strategic framing a_i is correlated with the regression error u_i, then the text feature $m(x_i)$ is *endogenous* — the familiar omitted-variable problem, now hidden inside the language.

Two Firms, Same Risk, Different Words

The endogeneity hazard, made concrete. Imagine two firms facing the *same* objective political risk s_i.

- Firm A's management discloses risks proactively — building a reputation for transparency.
- Firm B's management avoids politically sensitive topics — afraid of spooking the market.
- Same s_i, different strategic framing $a_i \Rightarrow$ different texts $x_i \Rightarrow$ different measured risk $m(x_i)$.

- If the disclosure choice a_i correlates with anything in u_i (governance quality, investor base, financial health), $m(x_i)$ is **endogenous** — before any fancy NLP even enters.

Setting from Hassan, Hollander, van Lent & Tahoun (2019, QJE).

EPU: The Validation Standard (Thread A)

Why is a word-count index taken seriously? Because every step of it was audited.

- **Human audit:** a trained team hand-coded 12,009 articles (1900–2012) under a 65-page guide; the six policy terms were selected from $\sim$32,000 candidate combinations to minimize false positives + false negatives.
- **Machine vs. human:** the "machine" is the automated word-count index itself — a program that flags every article matching the six-term rule and tallies the monthly counts, with no human judgment in the loop. Run over the same years the team hand-coded, it tracks the human-coded series at correlation 0.86 (quarterly, 1985–2012) and 0.93 (annual, 1900–2010); the gaps are uncorrelated with GDP growth.
- **Robustness:** dropping any one policy term barely moves the series; a 1,500-newspaper daily version correlates 0.85 with the 10-paper index.
- **And still — the identification critique:** does EPU *cause* downturns, or do downturns make journalists write "uncertainty"? The VAR ordering (EPU first) is a maintained assumption, not an identification argument.

Source: Baker, Bloom & Davis (2016, QJE).

Preview: The Machine Ahead

One equation will power everything in Acts 2–4

A Brief History: RNN → Transformer → GPT

- **2003:** Bengio et al. neural language model — continuous word vectors
- **2013:** Mikolov et al. *Word2Vec* — dense word embeddings at scale
- **2014:** Pennington et al. *GloVe* — global co-occurrence matrices
- **2017:** Vaswani et al. *"Attention Is All You Need"* — the Transformer replaces RNNs/LSTMs
- **2018:** Devlin et al. *BERT* (encoder); Radford et al. *GPT* (decoder)
- **2020–2022:** GPT-3, instruction tuning, RLHF → ChatGPT moment
- **2023–2026:** GPT-4/5, Claude (3–4), Gemini, LLaMA-3, DeepSeek; long-context (100k–1M tokens), open-weights ecosystem, agentic systems

Driver of the shift: attention scales better than recurrence, both in compute (parallel) and in quality (long-range dependencies).

The Autoregressive Principle

- An LLM is fundamentally a **next-token predictor**:

$$p\Big(t^{(n+1)} \mid t^{(1)}, t^{(2)}, \ldots, t^{(n)}\Big).$$

- **Generation:** sample $t^{(n+1)}$ from this distribution, append to the context, and repeat.
- **What this single equation enables** (the full revisit comes at the end of Act 4, once pretraining and fine-tuning are on the table):
  - **Summarizing 10-K filings or FOMC minutes** — prompt with document + instruction; the model predicts summary tokens one by one
  - **Generating narrative forecasts** conditional on macro state — condition on data in the prompt; next tokens extend the narrative
  - **Producing scenario analyses in plain language** — the scenario is the context; prediction fills in the implications
  - **Writing and debugging code** for empirical work — same objective over a code corpus yields code completions
- **Key insight:** "everything is next-token prediction" is mathematically tiny but pedagogically and practically enormous.